Kepler's Third Law and the Non-Material Medium

Cumulative Inflow, Atomic Cavities and a Common Origin of Weight and Inertia

Dmytro Kuzmenko

RWA® RAILWAYAUTOMATIC®

Email: info@ew2r3.org Website: Ew2R3.org

ORCID: 0009-0005-1975-1247

Ew2R3-15-2-1-EN-r17.pdf doi:10.13140/RG.2.2.34693.10721

© Dmytro Kuzmenko.

Abstract

We connect the orbital parameter $E = \omega^2 R^3$ to a hypothesis of a common origin of weight and inertial response. In this hypothesis a non-material medium has properties of an ideal incompressible fluid or gas and flows around impermeable atomic standing waves. Passage through an open toroidal centre is not entry into an atomic cavity. We associate the response with external-flow acceleration and changes in body motion relative to that flow. We distinguish the orbital law $U^2 R = E$ from the cumulative-front law $v^2 R^3 = C = E\ell^2$. We derive the equal-acceleration radius $R_* = \ell\sqrt{3/2}$ for the front and a released body. For $\ell = R_b$ the surface front acceleration is $1.5E/R_b^2$, while a body released from rest at $R_0 > 2R_b$ has a higher surface speed than the front. For the distinct spatial profile $u_r = -\sqrt{2E/R}$, the released body's speed remains below $|u_r|$. A local potential-flow comparison gives $\kappa = (1 + c_a)/(\beta + c_a)$: displaced volume cancels, but unit response additionally requires $\beta = 1$. We define geometric parameters for a controlled rotor comparison and conditions for testing a quantitative law. These analytical results do not determine the microscopic mechanism of atomic coupling. No new experimental measurements are performed.

Keywords: Kepler's third law, orbital parameter, non-material medium, E-field, cumulative flow, cavity collapse, weight, inertia, universality of free fall.

Contents

1 Introduction and place in the series **3**

2 Two kinematic processes **3**

2.1 The orbital invariant 3

2.2 The cumulative law 4

2.3 Known cumulative processes 4

3 Medium and matter as voids for each other **5**

3.1 Physical formulation 5

3.2 Calculations and physical assumptions 5

4 Levels of interaction description **5**

5 One coupling: weight and inertia **6**

5.1 Common physical formulation 6

5.2 Release of a body from rest 6

5.3 Kinematic comparison of inflow and body 7

5.4 Hydrodynamic analogies 8

6 Two regimes of relative motion and rotating devices **8**

7 Two motions in a vortex **9**

7.1 Medium flow and atomic-cavity motion 9

7.2 What is established for material two-phase flows 9

7.3 The whirlpool as a boundary between two media 10

7.4 Open question: the radial motion of atomic cavities 10

8 From circular to radial motion **10**

9 Tests and limits **10**

9.1 Universality of free fall 10

9.2 Rotor series 11

9.3 Conditions for testing and rejecting a quantitative law 11

9.4 Scope of this study 12

10 Conclusions **12**

Acknowledgments and declarations **12**

Data and code availability **13**

1 Introduction and place in the series

In the Ew2R3 research programme we use the orbital parameter $E = \omega^2 R^3$, determined from orbital lengths and periods. The first studies establish its relation to surface acceleration and characteristic speeds [1, 2] and propose a radial interpretation of free fall [3]. Here we connect these results to a hypothesis of matter and a non-material medium, with the aim of describing weight and inertial response through one coupling.

Companion studies treat components of multiple systems [4], the orbital-to-radial transition [5], compatibility between body motion and prescribed flow [6], and external flow around impermeable atoms [16]. This study connects these questions in one formulation. Each contribution has its own calculations and domain of applicability.

Main results. We connect the orbital and cumulative invariants through a length scale with consistent dimensions. We derive the radius at which the cumulative front and a released body have equal accelerations and the threshold for their surface speeds. We define rotor geometry parameters that can be separated in a controlled comparison. A local potential-flow comparison establishes the conditions under which displaced volume cancels from the response of a cavity. These results are analytical consequences of explicitly prescribed equations.

We distinguish four levels of the argument:

(a) physical assumptions about the medium and atomic cavities,

(b) quantitative results of earlier and companion studies,

(c) analytical relations derived here,

(d) undetermined parameters and experimental conditions for testing.

We present the hypothesis through its assumptions, equations and consequences. Dimensional consistency and trajectory calculations establish mathematical results. The physical mechanism of flow around atoms requires independent confirmation.

We characterize celestial sources by orbital parameters consistently with the first three studies [1, 2, 3]. The orbital calculations introduce neither celestial masses nor a universal gravitational constant. We use E-field as a name for the proposed non-material medium. In this study we do not identify it with the electromagnetic field. Their possible relationship requires a separate investigation. We describe coupling through acceleration response and changes in body motion relative to the flow, without assigning internal viscosity to the medium.

2 Two kinematic processes

2.1 The orbital invariant

For circular motion of radius R with angular speed ω and linear speed $U = \omega R$,

$$E = \omega^2 R^3 = U^2 R, \qquad [E] = L^3 T^{-2}. \tag{1}$$

The centripetal acceleration of that motion is $\omega^2 R = E/R^2$. This is the relation through which the earlier preprints obtain surface free-fall acceleration from orbits [1, 2]. For relative Keplerian ellipses the mean motion and semimajor axis replace ω and R [4].

2.2 The cumulative law

The second process is radial cumulative motion. The orbital-to-radial transition changes the physical regime, rather than merely replacing the symbol for speed. For a contracting front we consider a linear speed v increasing as $R^{-3/2}$. This differs from the orbital relation $U^2R = E$.

Because v is a linear speed, the product v^2R^3 has dimensions L^5T^{-2}, not those of E. We therefore write the cumulative law with its own constant,

$$v^2R^3 = C, \qquad C = E\,\ell^2, \qquad [C] = L^5T^{-2}, \tag{2}$$

where ℓ connects the normalization of the cumulative process to the orbital parameter of the same source. Thus E remains the source parameter, while ℓ supplies an additional length scale. Orbital E alone does not determine C. For the numerical comparison we set $\ell = R_b$, the source-body radius. The surface front speed then equals $\sqrt{E/R_b}$. This normalization is prescribed explicitly, rather than derived from the microscopic structure of the medium [5].

The two processes are summarized in Table 1.

Table 1: Two kinematic processes that must not be conflated.

	Orbital invariant	Cumulative law
Relation	$E = \omega^2R^3 = U^2R$	$v^2R^3 = C = E\ell^2$
Speed	angular ω or $U = \omega R$	linear v, radially inward
Radius	orbital radius, fixed on an orbit	radius of a contracting front
Behaviour as R decreases	U grows as $R^{-1/2}$	v grows as $R^{-3/2}$
Acceleration	$-E/R^2$ (centripetal)	$-3C/(2R^4)$ along the front
Dimensions of constant	L^3T^{-2}	L^5T^{-2}

The acceleration in the last column follows from Eq. (2) for a front that contracts while keeping v^2R^3 constant. With $v = -\dot{R}$,

$$\ddot{R} = \frac{1}{2}\frac{\mathrm{d}(\dot{R}^2)}{\mathrm{d}R} = -\frac{3C}{2R^4}. \tag{3}$$

The same expression is the radial material acceleration of a stationary profile $u_r = -\sqrt{C}\,r^{-3/2}$ [5]. That stationary profile, however, is not divergence-free in a spherical domain [5]. The cumulative law is therefore read here as the kinematics of a contracting front, not as a stationary incompressible field.

2.3 Known cumulative processes

The cumulative law has a well-known material counterpart. For an inviscid spherical cavity collapsing under a constant pressure difference Δp in a liquid of density ρ, released from rest at radius a_0, Rayleigh's first integral is [9, 5]

$$a^3\dot{a}^2 = \frac{2\Delta p}{3\rho}\left(a_0^3 - a^3\right). \tag{4}$$

The product $a^3\dot{a}^2$ approaches a constant only late in the collapse, when $a \ll a_0$. In that late phase the wall speed grows as $a^{-3/2}$, which is the cumulative law. Equation (4) also shows the physical meaning of the volumetric factor: the kinetic energy of the liquid set in motion by the wall is proportional to $\rho\,a^3\dot{a}^2$. This material example establishes compatibility of cumulative scaling with energy conservation under the stated conditions. It is not transferred to a celestial flow without a separate pressure law and boundary conditions.

3 Medium and matter as voids for each other

3.1 Physical formulation

In the proposed hypothesis we consider atoms as collapsed three-dimensional standing waves, bubble-like cavities that the non-material medium cannot enter. An aggregate of atoms creates a pressure difference for the medium. Under its own pressure the medium flows from regions of higher pressure to regions of lower pressure and around atoms on all sides, including through an open toroidal centre. Passage through that opening is not entry into the impermeable atomic cavity.

In the regime considered, the medium has the properties of an ideal incompressible fluid or gas with zero internal viscosity. For a fixed flux through a channel, continuity gives $Au = \text{const}$, so narrowing the available cross-section increases the local speed. We consider this external flow and its acceleration as the basis of coupling to a material body. Continuity alone does not determine the resultant force. A pressure field and boundary conditions are also required [16].

Opposite pressure trends. The hypothesis associates regions of high material-body pressure with reduced pressure of the medium. This correspondence sets the direction of the proposed inflow. Its quantitative dependence on body structure is not specified here.

Source and receiving body. The presence of a material body does not by itself establish its own gravitational flow. A source's own inflow must be identified separately from a body's response to an external flow, consistently with the multiple-system formulation [4].

Existing atoms and formation of matter. We distinguish coupling to existing impermeable atoms from the formation of standing waves. Cumulative conversion to an atomic state belongs to the broader physical hypothesis. Its microscopic mechanism is not derived here and is not replaced by absorption of the medium inside an existing atom.

3.2 Calculations and physical assumptions

The orbital relations of earlier studies [1, 2, 3] and the kinematic results here can be calculated without assuming that the microscopic structure of the atom has already been established. The standing wave, the impermeability of the atomic cavity and the relationship between pressures define their physical interpretation. The following sections calculate specific consequences and identify the additional parameters needed for a quantitative coupling law.

4 Levels of interaction description

We distinguish properties of the medium itself, coupling to atomic cavities and internal relations within a material body (Table 2). These three descriptive levels are not a list of fundamental forces. The accepted physical classification distinguishes gravitational, electromagnetic, strong and weak interactions [17]. We do not derive a new classification or compare the magnitude of interatomic forces to the proposed medium response.

Absence of internal viscosity does not imply absence of a reaction to body acceleration. In the baseline regime we require no braking of uniform motion through a homogeneous medium that is relatively stationary in the chosen local frame. A positive term proportional to the velocity difference would produce decay in this regime and would need separate justification [16]. A conclusion about changes of actual orbits would require a complete flow model and a quantitative comparison with observations.

Table 2: Descriptive levels in the problem formulation.

Level	Physical formulation	Treatment here
(i) medium – medium	Zero internal viscosity in the stated regime	Ideal incompressible fluid/gas properties
(ii) medium – atomic cavities	External flow and body response	Acceleration, weight and changes of motion relative to the flow
(iii) atom – atom	Bonding and collective body structure	No microscopic law or quantitative force comparison is specified here

5 One coupling: weight and inertia

5.1 Common physical formulation

We consider free fall, the weight of a supported body and inertial response as regimes of one coupling between atomic cavities and external flow. The medium moves between atoms, around their boundaries and through open toroidal centres. It does not enter the atomic cavities themselves.

In free fall the body is not held by a support. When a support prevents that motion, it receives the reaction measured as weight. A change in body motion relative to the medium, including a change of direction, produces an inertial response. The hypothesis connects both reactions to external flow around atoms rather than internal viscosity of the medium.

The relations $v_{\text{front}}^2 = C/R^3$ and $a_{\text{body}} = -E/R^2$ describe different kinematic processes. Their difference is not itself a friction coefficient, since squared speed and acceleration have different dimensions. We therefore compare accelerations and speeds separately and use a conditional response law [16]. Equality of weight and inertial coefficients must follow from that law under matching conditions, rather than from a common name.

5.2 Release of a body from rest

For a stationary central source, neglecting air and rotation, prescribe $\ddot{R} = -E/R^2$, $R(0) = R_0$ and $\dot{R}(0) = 0$. Integration gives

$$v_{\text{body}}^2 = 2E\left(\frac{1}{R} - \frac{1}{R_0}\right). \tag{5}$$

This calculates the trajectory for a prescribed acceleration. In the local potential-flow comparison it corresponds to unit response $\kappa = 1$ without additional velocity damping [16].

For the comparison spatial profile $u_r = -\sqrt{2E/R}$, the material acceleration equals $-E/R^2$. Consequently,

$$\frac{v_{\text{body}}}{|u_r|} = \sqrt{1 - \frac{R}{R_0}} < 1 \qquad (0 < R < R_0). \tag{6}$$

This inequality applies to this profile and these initial conditions. The spatial profile u_r and the cumulative front $v_{\text{front}} = \sqrt{E\ell^2/R^3}$ are different descriptions. Equation (6) does not apply to the latter. Continuity of the spatial profile requires separate treatment of the phase balance [16].

For release from 1 km above Earth, $E = 3.986004418 \times 10^{14}\ \text{m}^3\text{s}^{-2}$ and $R_b = 6371$ km, the fall time is 14.273 s and the surface speed is 140.134 $\text{m}\,\text{s}^{-1}$. For $\ell = R_b$ the surface cumulative-front speed is 7909.79 $\text{m}\,\text{s}^{-1}$, while $|u_r| = 11186.14\ \text{m}\,\text{s}^{-1}$. In this particular example the body speed is below both comparison speeds. This does not establish that ordering for an arbitrary release height.

Compatibility of body motion with a prescribed flow is studied separately [6]. Corotation, zero tangential slip and radial fall are distinct conditions [5]. Equality of weight and inertial coefficients for real bodies of different composition requires independent determination of the response parameters.

5.3 Kinematic comparison of inflow and body

We compare two prescribed laws. To compare like with like, we use accelerations, both with dimensions LT^{-2}: the acceleration of the cumulative front, Eq. (3), and the acceleration of a body released from rest,

$$a_{\mathrm{flow}}(R) = -\frac{3C}{2R^4} = -\frac{3E\ell^2}{2R^4}, \qquad a_{\mathrm{body}}(R) = -\frac{E}{R^2}. \tag{7}$$

Their difference is

$$\Delta a(R) = a_{\mathrm{flow}} - a_{\mathrm{body}} = -\frac{E}{R^2}\left[\frac{3\ell^2}{2R^2} - 1\right], \qquad \frac{a_{\mathrm{flow}}}{a_{\mathrm{body}}} = \frac{3}{2}\left(\frac{\ell}{R}\right)^2. \tag{8}$$

The two accelerations are equal at

$$R_* = \ell\sqrt{3/2} \simeq 1.224745\,\ell. \tag{9}$$

Inside R_* the front acceleration has a larger magnitude than the free-body acceleration. Front acceleration scales as R^{-4}, while body acceleration scales as R^{-2}. Outside R_* it accelerates less. Under the surface normalization $\ell = R_b$, the cumulative acceleration at the surface is $1.5\,E/R_b^2$ and the equality radius lies at $1.224745\,R_b$. For the Earth, with the parameter quoted above and without rotation, $E/R_b^2 = 9.8203\ \mathrm{m\,s^{-2}}$, $a_{\mathrm{flow}}(R_b) = 14.730\ \mathrm{m\,s^{-2}}$, and $R_* = 7802.85$ km, which is 1431.85 km above the surface.

The same comparison can be made for speeds, again of equal dimensions. A body released from rest at R_0 has $v_{\mathrm{body}}^2 = 2E(1/R - 1/R_0)$, while the cumulative front has $v_{\mathrm{flow}}^2 = E\ell^2/R^3$. With $\ell = R_b$, at the surface

$$\left.\frac{v_{\mathrm{body}}^2}{v_{\mathrm{flow}}^2}\right|_{R_b} = 2\left(1 - \frac{R_b}{R_0}\right). \tag{10}$$

For $\ell = R_b$ and $R_b < R_0 < 2R_b$, the body reaches the surface with a lower speed than the cumulative front. For $R_0 > 2R_b$ it reaches the surface with a higher speed than that front. These are exact kinematic orderings of the two prescribed speeds, not of arrival times. In the proposed interpretation they are the two regimes named by the author: inside the threshold the medium moves through the body, between its impermeable atoms and through open toroidal centres, and outside it the body moves through the medium. Both regimes must be described by one coupling.

This comparison concerns the front speed and the body speed at the same radius. Reading it as local relative motion additionally requires a spatial velocity field for the medium and a relation between that field and the front speed [5, 6].

These relations are elementary consequences of the two laws and of the chosen normalization. They do not supply the coupling law. They identify where its two regimes meet and what any proposed law must reproduce. The value $\ell = R_b$ is an illustrative choice. A different physically specified ℓ moves R_* proportionally, and also changes the release threshold. Equating the two speeds at the surface for general ℓ gives

$$R_0 = \frac{R_b}{1 - \ell^2/(2R_b^2)}, \tag{11}$$

which is finite only for $\ell^2 < 2R_b^2$: for $\ell = 0.5\,R_b$ it is $1.142857\,R_b$, for $\ell = R_b$ it is $2R_b$, and for $\ell^2 \geq 2R_b^2$ no body released from rest reaches the surface with a higher speed than the cumulative front.

5.4 Hydrodynamic analogies

For comparison we use classical hydrodynamics of an ideal fluid with explicit conditions on potentiality, local uniformity and cavity shape. This checks the response structure without identifying a classical bubble with an atomic standing wave.

No braking, but a response to changes of motion. In steady potential flow of an inviscid, incompressible fluid, a body moving uniformly experiences no resistance. When its motion relative to the fluid is changed, it experiences an additional reaction proportional to the relative acceleration, the added-mass effect [11, 10]. For a sphere the added mass equals half of the fluid displaced [11]. Uniform motion is therefore not braked, but every change of motion is resisted. These statements hold for potential flow and its stated boundary conditions. Separated, viscous or rotational flows add further terms.

Volumetric scaling of response. In the local potential-flow comparison, both the pressure-gradient reaction of an accelerating flow and the added-inertia reaction are proportional to displaced volume. The effective response for a spherical comparison is

$$\kappa = \frac{1 + c_a}{\beta + c_a}, \qquad \beta + c_a > 0, \tag{12}$$

where c_a describes added inertia and β is the intrinsic effective inertial coefficient relative to displaced-medium inertia [16]. Volume cancels for fixed β and c_a. For a sphere $c_a = 1/2$, so $\beta = 0$ gives $\kappa = 3$, whereas unit response requires $\beta = 1$. Independence of acceleration from volume and unit response are therefore distinct conditions. We do not assign an atomic standing wave the zero intrinsic inertia of an empty classical bubble. A microscopic justification of $\beta = 1$ is not obtained here.

Interaction of cavities through the medium. Two pulsating bodies in an ideal fluid act on each other through the fluid alone: when they pulsate in phase the fluid pushes them toward each other, in antiphase it pushes them apart, and this mutual action decreases with the inverse square of their separation [12]. This result of the Bjerknes programme was developed in the nineteenth and early twentieth centuries as a hydrodynamic analogy of electric and magnetic fields [12]. It is a historical precedent for an inverse-square action between cavities that is mediated entirely by a medium and contains no mass parameter of the bodies. It is cited here as a precedent. The present model does not assume pulsating atoms.

These are analogies, not demonstrations. They show that the combination "no braking of uniform motion, but a response to changes of motion", and an action between cavities carried by the medium, are consistent with an ideal medium. The analogies also do not by themselves remove a dependence on shape, on interactions between neighbouring cavities, or on composition. If the hypothesis is correct, these dependences must cancel to the precision with which the universality of free fall has been measured (Section 9).

The companion study of external flow around impermeable atoms [16] formulates the pressure-gradient response, the role of effective standing-wave inertia and a possible small relative-velocity correction. A permeable point sink is only a classical comparison, not the atomic boundary condition. The principal acceleration response remains present when that additional correction vanishes.

6 Two regimes of relative motion and rotating devices

We consider flow of the medium relative to a material body and motion of a body through the medium. The latter includes rotating instruments and changes in the direction of motion of their atoms. Rotor parameters should distinguish rim kinematics from a possible geometric dependence of external flow. Precession alone does not establish a new medium effect.

The corresponding quantities are defined as follows. For a rotor of radius r and angular speed ω, with rim speed $U = \omega r$,

$$E_{\text{rot}} = \omega^2 r^3, \qquad C_{\text{rot}} = U^2 V_{\text{eff}}, \qquad E_{\text{coupled}} = \frac{C_{\text{rot}}}{A_{\text{eff}}} = \omega^2 r^2 \, \frac{V_{\text{eff}}}{A_{\text{eff}}}, \tag{13}$$

where V_{eff} is the volume set in motion and A_{eff} the area through which the rotor exchanges motion with the medium. E_{rot} is the orbital-type invariant, Eq. (1), while C_{rot} is the volumetric, cumulative-type quantity. And E_{coupled} brings the latter back to the dimensions of E through the length $V_{\text{eff}}/A_{\text{eff}}$. The ratio

$$\frac{E_{\text{coupled}}}{E_{\text{rot}}} = \frac{V_{\text{eff}}}{A_{\text{eff}}\, r} \tag{14}$$

depends on the prescribed effective volume and area. It is a geometric parameter if these equal geometric volume and area. For a sphere it is 1/3. For a thin disc of thickness d, taking both faces as the exchange area and neglecting the rim, it is $d/(2r)$. Two discs of the same radius $r = 5$ cm and thicknesses 1 and 2 cm therefore have equal E_{rot} at equal ω, but values of E_{coupled} in the ratio 1:2. The 1:2 ratio applies specifically to the two-face area choice. Including the rim gives total area $2\pi r(r + d)$ and geometric ratio $d/[2(r + d)]$. For these discs the values are 1/12 and 1/7, with ratio 12/7. For geometrically similar rotors both parameters are proportional when effective geometry is defined consistently. For rotors of equal radius and different thickness it becomes measurable. This provides the experimental lever described in Section 9.

The classical description of rigid-body rotation and precession is used here only as the established control description, whose prediction must be subtracted from any measurement. It is not the basis of the model: within the hypothesis, the inertial response of the rotor is itself a consequence of the coupling between its atomic cavities and the medium.

7 Two motions in a vortex

7.1 Medium flow and atomic-cavity motion

We distinguish medium flow from motion of material atoms in a vortex. In the hypothesis, the medium flows into the vortex that it spins up, while atomic cavities are entrained in circular motion and can migrate radially in the opposite direction. The direction and speed of migration must be determined from the response law, rather than the kinematic picture alone.

The orbital parameter supplies a common quantitative starting point for comparing atomic to galactic scales [2]. A common parameter form does not establish a common mechanism for every object. We use material two-phase flows as comparisons to distinguish circular entrainment from radial migration.

7.2 What is established for material two-phase flows

In ordinary fluids the motion of a dispersed phase in a vortex is well documented, and it depends on the density of that phase relative to the carrier.

- Light inclusions move toward the vortex axis. Gas nuclei and bubbles released near a line vortex are drawn into its low-pressure core by the radial pressure gradient. This capture is predicted by point-particle models and observed experimentally [13]. Small bubbles are used routinely to make vortex cores visible for the same reason.
- Heavy inclusions are expelled from vortical regions. Particles denser than the carrier lag behind the flow and collect in regions of low vorticity and high strain, the effect known as preferential concentration [14].

- In both cases the inclusions are carried around by the rotating flow while they migrate radially, so that circular entrainment and radial separation occur together.

These observations demonstrate that the carrier and its inclusions be described as two different motions in the same space, with radial separation and circular entrainment acting simultaneously. They also show that, in material flows, the direction of the radial migration is fixed by the density contrast: light cavities go to the axis, heavy particles go outward.

7.3 The whirlpool as a boundary between two media

A whirlpool illustrates coexistence of radial and circular motion at the boundary of two material media. In an idealized free vortex of constant circulation outside its core, tangential speed decreases as $1/r$, whereas Keplerian speed in Eq. (1) decreases as $r^{-1/2}$. A whirlpool free surface and three-dimensional celestial inflow impose different boundary conditions. This is a comparison of motion structure, rather than identity of profiles or measurement of a non-material medium.

7.4 Open question: the radial motion of atomic cavities

Capture of material bubbles toward a vortex core does not determine the direction of motion of atomic standing waves. In a classical flow that direction depends on density contrast and the pressure law. Atomic cavities require corresponding effective parameters and boundary conditions. In our formulation an orbit must reconcile circular entrainment, radial inflow action and the response of atomic structure. Section 5 establishes kinematic conditions for that reconciliation. Those conditions do not yet determine a quantitative radial-migration law.

8 From circular to radial motion

We treat the orbital-to-radial transition as a change of physical regime. The cumulative law specifies front speed in a radial phase, while the orbital invariant specifies circular kinematics. Both use the same source parameter E, but the cumulative law additionally requires the length ℓ.

Corotation alone does not force a body to fall. The companion calculation checks continuity of a three-dimensional flow and constrains the cubic radial tendency [5]. A separate calculation formulates release and comoving conditions for a specified coupling class [6]. Equations (8)–(9) establish equality of two accelerations, rather than a transition law between them.

Joint radial motion of body and medium with zero velocity difference and release of a body from rest are different initial conditions. Connecting them requires the evolution of the flow and the body response. Replacing angular speed with linear speed does not provide that evolution.

9 Tests and limits

9.1 Universality of free fall

The response law must be consistent with measured universality of free fall. MICROSCOPE compared titanium and platinum alloys and obtained $\eta = [-1.5 \pm 2.3(\text{stat}) \pm 1.5(\text{syst})] \times 10^{-15}$ without detecting a violation of the weak equivalence principle [15]. The statistical uncertainty is quoted at one standard deviation. This is a result for a specific comparison, rather than a universal sharp 10^{-15} cutoff for all materials and experiments.

If both test bodies satisfy $a_i = \kappa_i a_{\mathrm{m}}$ for a common external acceleration, a quantitative law predicts

$$\eta_{\mathrm{pred}} = \frac{2(\kappa_{\mathrm{Ti}} - \kappa_{\mathrm{Pt}})}{\kappa_{\mathrm{Ti}} + \kappa_{\mathrm{Pt}}}. \tag{15}$$

This prediction must be compared using the sign convention, measurement conditions and statistical and systematic uncertainties of the published analysis. Equal κ_i give zero regardless of their common value. Relative-fall testing alone therefore does not determine the absolute response coefficient.

9.2 Rotor series

Separating the parameters in Eq. (13) requires independent changes of geometry, material and speed. Comparisons can use rotors of equal radius and different thickness, equal geometry and different material, and equal material with different radii. Each measurement records rotation speed, support reaction, precession, vibration and temperature. Controls must account for ordinary inertia, torque, supports, air and the measuring apparatus dynamics.

Before measurement, specify a law for the observable residual $\delta = \Delta W/W_0$, effective V_{eff} and A_{eff}, and a predicted effect size. Different geometric parameters alone do not predict a weight change. This study defines comparison parameters, rather than reporting a completed rotor experiment.

9.3 Conditions for testing and rejecting a quantitative law

The following conditions constitute a test programme. No new laboratory measurements are performed here. Rejecting a particular law requires its parameters, domain, numerical prediction and uncertainty comparison rule to be specified in advance.

1. **Rotor residual.** If a law predicts a nonzero reaction resolvable by the apparatus, and the controlled result with stated uncertainty is incompatible with it, that law with its specified parameters is rejected. Without such a prediction a null residual supplies only an upper limit on the effect.

2. **Geometry dependence.** If a specified law predicts separated reactions at equal E_{rot} and different $V_{\mathrm{eff}}/A_{\mathrm{eff}}$, and a sufficiently sensitive experiment excludes that separation, that dependence is rejected. Curve coincidence without a specified prediction does not establish that volume is generally irrelevant.

3. **Test-body composition.** Compare Eq. (15) for the corresponding titanium and platinum samples with MICROSCOPE and its uncertainties at a chosen confidence level. Incompatibility excludes the tested form of composition dependence. A universal single-number 10^{-15} cutoff is not applied.

4. **Weight and inertial response.** If the same law predicts equality of independently determined coefficients, compare their difference under matching conditions, consistent normalization and joint uncertainty. Significant disagreement rejects that quantitative form, rather than all possible common-origin mechanisms.

5. **Uniform motion.** In the baseline regime of a homogeneous, locally stationary medium without external forces, require $\dot{\mathbf{v}} = 0$ even when $\mathbf{v} - \mathbf{u} \neq 0$. A law that damps this motion does not satisfy the baseline. Constraints from actual orbits separately require a flow field and a prediction for trajectory changes.

Conditions 1–4 require numerical specification and measurements. Condition 5 can be checked as a limiting case of a prescribed equation. Nonconfirmation of an effect, a bound on its magnitude and rejection of a specified law are different outcomes.

9.4 Scope of this study

The kinematic relations follow from prescribed laws. The normalization $\ell = R_b$ is illustrative. The potential-flow comparison determines how κ depends on effective coefficients, but does not derive them from an atomic standing wave. Atom formation, a quantitative relation between the two pressures and microscopic interatomic forces are not calculated. No new measurement of a non-material flow is claimed.

10 Conclusions

Physical picture. We propose one non-material medium with the properties of an ideal incompressible fluid or gas. Under its own pressure it flows from regions of higher to regions of lower pressure, toward aggregates of atoms, around impermeable atomic standing waves and through open toroidal centres. We propose that the weight of a supported body and the inertial response to a change of its motion relative to the flow are two manifestations of this one external coupling.

Two processes, one source parameter. We establish that the orbital law $U^2R = E$ and the cumulative law $v^2R^3 = C = E\ell^2$ describe distinct processes with constants of different dimensions, connected through one length ℓ. The same source parameter E characterizes both.

Where the two regimes meet. We derive the radius $R_* = \ell\sqrt{3/2}$ at which the cumulative front and a body released from rest have equal accelerations. For $\ell = R_b$ the surface front acceleration is $1.5E/R_b^2$, and a body released beyond $2R_b$ reaches the surface with a higher speed than the front, while for $R_b < R_0 < 2R_b$ its surface speed is lower. In the proposed interpretation these are the author's two regimes: the medium moving through the body between its impermeable atoms, and the body moving through the medium. The comparison does not identify the front speed with the local medium velocity. For general ℓ the threshold is Eq. (11), finite for $\ell^2 < 2R_b^2$. In the spatial profile $u_r = -\sqrt{2E/R}$ a released body approaches the medium speed as $\sqrt{1 - R/R_0}$ [16].

One coefficient for weight and inertia. In the local potential-flow comparison the pressure-gradient force of an accelerating medium and the added-inertia response both scale with displaced volume, so that $\kappa = (1 + c_a)/(\beta + c_a)$: for fixed β and c_a the common volume factor cancels, and in the hypothesis the remaining coefficient β is a property of the standing wave of the medium rather than a mass parameter. At $\beta = 1$ the response is exactly unity, and weight and inertial response follow from one coefficient.

Tests. At equal ω, rotors of equal radius and different thickness separate E_{rot} from E_{coupled}. Composition universality is expressed through η_{pred} and compared with MICROSCOPE [15].

Next step. The coefficient β of the atomic standing wave and the pressure law of the medium remain to be derived. With them, the rotor and composition tests become quantitative predictions.

Acknowledgments and declarations

The author thanks RWA® RAILWAYAUTOMATIC® for their support of this research.

Funding. This work was funded solely by RWA® RAILWAYAUTOMATIC®, which has self-funded the Ew2R3 research programme since 2013. No other funding was received.

Competing interests. The author is affiliated with RWA® RAILWAYAUTOMATIC®, which supports the research programme. The author declares no other competing interests.

AI-based tools were used to assist with calculations, code development and manuscript preparation. The author takes full responsibility for the final content.

Data and code availability

An earlier Ew2R3 visualization release is archived on Zenodo [7]. Calculations, input data and interactive visualization for this study are part of the Ew2R3 software release [8]. The programme website is Ew2R3.org.

No new observational data were generated. All numerical values in Sections 5 and 6 follow from the formulas given in the text and the stated input values.

References

[1] D. Kuzmenko, *The Gravitational Parameter $E = \omega^2 R^3$ as the Primary Constant of a Celestial System: Surface Gravity from Orbital Observations Without Masses or G*. ResearchGate preprint, doi:10.13140/RG.2.2.10828.37764.

[2] D. Kuzmenko, *The Orbital Invariant $E = \omega^2 a^3$ Across Scales: From the Hydrogen Atom to the Milky Way*. ResearchGate preprint, doi:10.13140/RG.2.2.24728.56327.

[3] D. Kuzmenko, *Radial Flow Interpretation of the Gravitational Invariant $E = \omega^2 R^3$: A Hydrodynamic Model for Free-Fall Acceleration Without Mass*. ResearchGate preprint, doi:10.13140/RG.2.2.33012.31364.

[4] D. Kuzmenko, *Kepler's Third Law in Multiple Systems: Component Orbital Parameters and Controlled Group Approximations*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.12882.72646.

[5] D. Kuzmenko, *Kepler's Third Law and the Orbital-to-Radial Flow Hypothesis: Corotation, Continuity and Moving Boundaries*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.26304.49927.

[6] D. Kuzmenko, *Kepler's Third Law and Body–Flow Coupling: A Cavity Hypothesis and Conditional Compatibility Tests*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.19593.61282.

[7] D. Kuzmenko, *Ew2R3.org interactive visualization: $E = \omega^2 R^3$*, software, v0.1.0-rc.3 (2026). Zenodo. Release doi:10.5281/zenodo.21762453. Project doi:10.5281/zenodo.21757888. Website: Ew2R3.org.

[8] D. Kuzmenko, *Ew2R3 shared calculations and interactive visualization: matter, multiple systems, orbital-to-radial transition and body–flow coupling*. Computer software and data (2026) doi:10.5281/zenodo.23627938.

[9] Lord Rayleigh, On the pressure developed in a liquid during the collapse of a spherical cavity, *Philosophical Magazine* 34, 94–98 (1917). doi:10.1080/14786440808635681.

[10] M.R. Maxey and J.J. Riley, Equation of motion for a small rigid sphere in a nonuniform flow, *Physics of Fluids* 26, 883–889 (1983). doi:10.1063/1.864230.

[11] C.E. Brennen, *Cavitation and Bubble Dynamics*, Oxford University Press, New York (1995), chapter 5.

[12] V. Bjerknes, *Fields of Force*, Columbia University Press, New York (1906).

[13] G.F. Oweis, I.E. van der Hout, C. Iyer, G. Tryggvason and S.L. Ceccio, Capture and inception of bubbles near line vortices, *Physics of Fluids* 17, 022105 (2005). doi:10.1063/1.1834916.

[14] K.D. Squires and J.K. Eaton, Preferential concentration of particles by turbulence, *Physics of Fluids A* 3, 1169–1178 (1991). doi:10.1063/1.858045.

[15] P. Touboul et al., MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle, *Physical Review Letters* 129, 121102 (2022). doi:10.1103/PhysRevLett.129.121102, arXiv:2209.15487.

[16] D. Kuzmenko, *Kepler's Third Law and the Flow of an Ideal Medium Around Atoms*. ResearchGate preprint (2026) doi:10.13140/RG.2.2.33015.38562.

[17] CERN, *The Standard Model*. https://home.web.cern.ch/science/physics/standard-model. Accessed 10 October 2026.

[18] D. Kuzmenko, *$E{=}\omega^2R^3$: Kepler's Third Law as the Basis of a Unified Physical Picture: Results of the Ew2R3.org Programme and a Call for Independent Verification*. ResearchGate preprint (2026). doi:10.13140/RG.2.2.24393.30561.